

Project MediLong

Frontend & UI/UX Squad — Detailed Report

Squad Report · 27-Day Sprint · Sept 2 – Sept 28

Squad	Frontend & UI/UX
Members	Maadhesh, Sanjeevram, Adithyan, Kishore
Sprint Window	Sept 2 – Sept 28, 2026 (27-day sprint)

Mission

Build the multi-step clinical intake flow, the interactive SHAP explanation widgets, and the PDF export preview — the only surface the demo audience actually sees.

Contract Slice Owned

Consumes the full unified JSON contract (all 4 top-level objects) once populated by Backend; before that, renders against a locally mocked copy of the same shape so work is never blocked.

Sprint-by-Sprint Tasks

Sprint 1 (Sept 2–8)

- Scaffold app (React/Next.js or Streamlit — lock choice by Sept 3).
- Build 3-pillar intake UI shell: demographics form, ultrasound image upload, lab PDF upload.
- Wire screens to a local mock JSON matching the unified contract exactly — no live calls yet.
- Screen routing: Upload → Review → Predict → Explain.

Sprint 2 (Sept 9–15)

- SHAP waterfall/bar chart component (feature name + impact direction/magnitude).
- Editable-field review screen: user can correct OCR-extracted biomarkers before submission (source_confidence drives which fields are flagged).
- PDF export preview pane.

Sprint 3 (Sept 16–22)

- Swap all mock calls for live Backend endpoints.
- Handle real latency/loading states and error responses (e.g. failed OCR, corrupted image).
- Display stub-mode disclosure for Pillar 2 results in the UI (mode: "stub").

Sprint 4 (Sept 23–28)

- Edge-case UI states: partial uploads, missing biomarkers, safety-override alert banner (e.g. hypothyroidism differential).
- Visual polish pass and demo rehearsal walkthrough.

Dependencies

- Blocked by: Backend (live endpoints), ML Fusion (SHAP payload shape).
- Blocks: nothing downstream — Frontend is the terminal consumer.

Definition of Done

- All 4 screens functional against live data with zero remaining mock dependencies.
- Stub-mode clearly disclosed to the user for Pillar 2.
- Guardrail alerts visually distinct from routine risk output.